

Results Report Phase T — Inductive Miner (PM4Py)

Aggregated from the stored ratings of the questionnaire prototype (`.miner_cache/phase_t_eval`, `.miner_cache/phase_e_eval`) and the metrics of the runs (`.miner_cache/results/.../imp__7f31529c/result_data.json`). Format follows the example reports of the Heuristics Miner, kept more concise.

Answers drawn from the characterization literature of the Inductive Miner (Leemans et al. 2013; 2014) and from the tooling documentation of the chosen setup (Berti et al. 2023, PM4Py).

Structured

Item	Answer	Rationale
T-S-BQ-1 Active restriction beyond the recorded traces, rather than mere log reproduction?	Yes	Cut detection actively restricts behavior (seed, not overridden).
T-S-BQ-2 Documented guarantee that the observed variants remain replayable?	Yes	IM keeps every variant replayable; noise is a filter, not a replay loss (seed).
T-S-SF-1 Explicit, connected control-flow structure or losslessly equivalent notation?	Yes	Process tree / Petri net: explicit connected control flow with start/end (seed).

Semi-Structured

Item	Answer	Rationale
T-Sm-BQ-1 Can it enforce fixed local orderings on part of the behavior?	Yes	Sequence cuts of the process tree fix local ordering on sub-regions, not a uniform level of restriction.
T-Sm-BQ-2 Can it leave part of the behavior open/unordered?	Yes	Parallel (AND) cuts leave parts of the ordering open and unordered.
T-Sm-SF-1 Fixed and open dependencies side by side in one model?	No	Evaluator answer No, deviating from the seed (n/a, "single-mode imperative").

Loosely Structured

Item	Answer	Rationale
T-L-BQ-1 Evidence-driven individual decision per restriction (only with sufficient log support)?	No	IM decides via recursive cuts with a global noise filter, not per restriction individually.
T-L-BQ-2 Documented generalization, so that the output accepts behavior beyond the exact traces?	No	Evaluator answer No, deviating from the seed (Yes, "cuts/concurrency inference").
T-L-SF-1 Binding via local constraints rather than a fixed path structure?	No	The imperative output requires a connected path structure (seed).

Tfit Result

Structuredness class	#Yes / #Items	Tfit (%)
Structured	3/3	100.0
Semi-Structured	2/3	66.7
Loosely Structured	0/3	0.0

Interpretation. The a-priori profile is skewed toward the structured side (100% / 66.7% / 0%). The documented mechanisms of the IM are consistently imperative: it reconstructs the filtered behavior into a connected process tree or Petri net and guarantees replayability. On the semi middle, the process tree a-priori covers both reframed capabilities — sequence cuts fix local ordering (Sm-BQ-1), parallel cuts leave parts open (Sm-BQ-2); only the representation of fixed and open bindings side by side in one model (Sm-SF-1) is missing. At the loose pole the profile drops to 0%: the IM does not decide in an evidence-driven way per restriction (L-BQ-1, new question wording → No), does not, according to the evaluator, generalize beyond the variants (L-BQ-2), and offers no constraint-based notation (L-SF-1). Note that this describes the a-priori capability of the documented mechanisms; the empirical output (Phase E) tests what actually happens on the concrete logs — there, the semi middle remains the hardest class (Efit 0%).

Note on the setup. The Phase E runs use the PM4Py default setup with noise threshold 0.0 (classic IM), while the Phase T seeds partly refer to the IMf default (0.20). The difference is immaterial for the a-priori capability, but is carried along for the empirical interpretation.

Sources. Leemans, S. J. J., Fahland, D. & van der Aalst, W. M. P. (2013): Discovering Block-structured Process Models from Event Logs. · Leemans et al. (2014): Discovering Block-structured Process Models from Event Logs Containing Infrequent Behaviour. · Berti, A., van Zelst, S. & Schuster, D. (2023): PM4Py.

Results Report — Phase E (empirical), Inductive Miner · all three structuredness classes · BQ & IN

1 Objective and procedure

The models discovered with the Inductive Miner (PM4Py, `discover_process_tree_inductive + convert_to_petri_net`, noise 0.0) are evaluated against the empirical Phase E items of all three classes. Four items per class: two on Behavioral Quality (BQ), two on Interpretability (IN). Three-point scale per item (2 = full, 1 = partial, 0 = absent). Each item is averaged over the three logs of the class, then Efit = sum of item means / 8 × 100. Phase T later enters the overall fit with 3/7·Tfit, Phase E with 4/7·Efit.

Note on fitness. Token-replay fitness is 1.000 for all nine models (100% fitting traces), because the IM by construction keeps every variant replayable. For the replayability of the observed variants (BQ-2) the value remains meaningful; the precision, however, shows that the allowed behavior on Semi/Loosely is massively inflated by broad AND or skip constructs (precision 0.34–0.88). The precision side of the BQ assessment (BQ-1) therefore reads precision and model form, not fitness alone.

2 Data basis: metrics of all models

Log	Cl.	Fitness	Fit-Traces	Precision	Silent	Edges	CFC	Simpl.
Log01_structured	S	1.000	100%	1.000	2	18	3	0.895
Log04_structured	S	1.000	100%	1.000	0	14	3	0.867
Log05_structured	S	1.000	100%	0.953	0	12	6	0.714
Log02_semiStructured	Sm	1.000	100%	0.486	3	26	5	0.677
Log06_semiStructured	Sm	1.000	100%	0.342	3	32	3	0.684
Log07_semiStructured	Sm	1.000	100%	0.446	2	32	5	0.684
Log03_looselyStructured	L	1.000	100%	0.744	8	34	12	0.619

Log	Cl.	Fitness	Fit-Traces	Precision	Silent	Edges	CFC	Simpl.
Log08_looselyStructured	L	1.000	100%	0.879	9	36	13	0.636
Log11_looselyStructured	L	1.000	100%	0.543	5	22	10	0.630

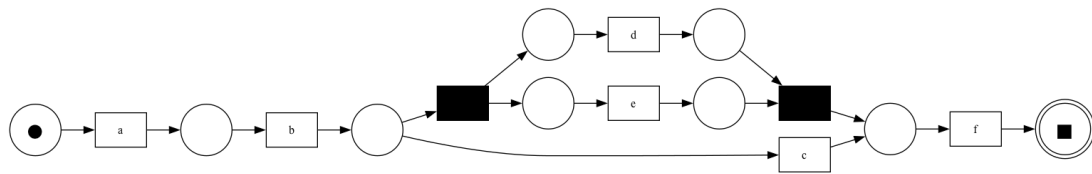
CFC = control-flow complexity, Simpl. = simplicity (arc-degree). Silent = τ -transitions from the process-tree-to-Petri-net mapping.

3 Class Structured — Efit = 95.8%

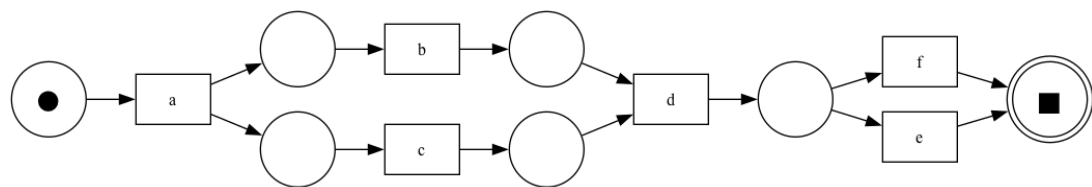
Logs: Log01, Log04, Log05. Behavioral Quality 91.7% · Interpretability 100.0%.

Item	Challenge	Log01	Log04	Log05	Mean	Question (short)
E-S-BQ-1	Behav. Quality	2	1	2	1.67	How tightly does the output bound the behavior space?
E-S-BQ-2	Behav. Quality	2	2	2	2.00	Do the attested variants remain replayable?
E-S-IN-1	Interpret.	2	2	2	2.00	How compactly does the output represent the behavior?
E-S-IN-2	Interpret.	2	2	2	2.00	Can the ordering be read off from the output?
Efit sum					7.67/8	= 95.8%

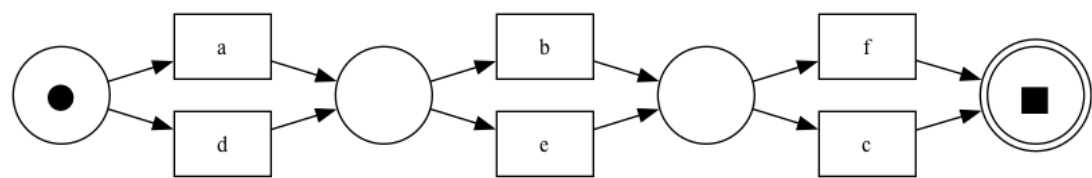
Rationale per item. BQ-1: precision 1.000 / 1.000 / 0.953, clear control flow without unnecessary optionality; Log04 rated marginally tighter (1). BQ-2: fitness 1.000, all variants replayable → all 2 (no downgrade for Log04 needed). IN-1: only as many elements as necessary, very compact (Simpl. 0.71–0.90, CFC 3–6) → all 2. IN-2: clear left-to-right ordering, τ -transitions are pure AND-split/joins and do not impair readability → all 2.



Log01 · Structured · Petri net



Log04 · Structured · Petri net



Log05 · Structured · Petri net

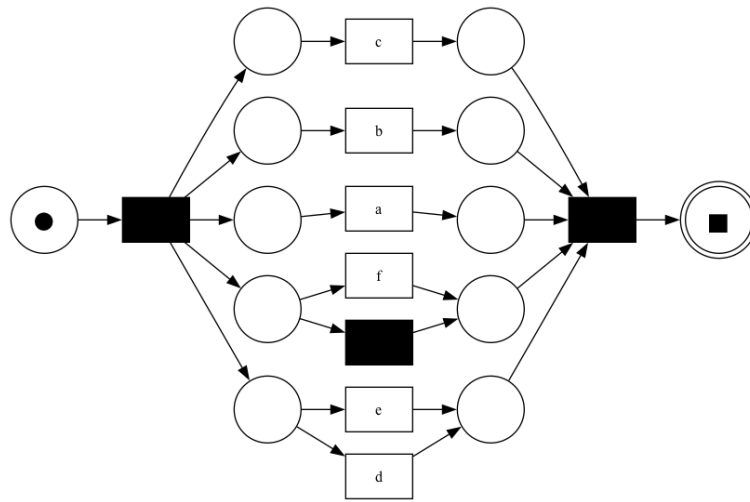
4 Class Semi-Structured — Efit = 12.5%

Logs: Log02, Log06, Log07. Behavioral Quality 25.0% · Interpretability 0.0%.

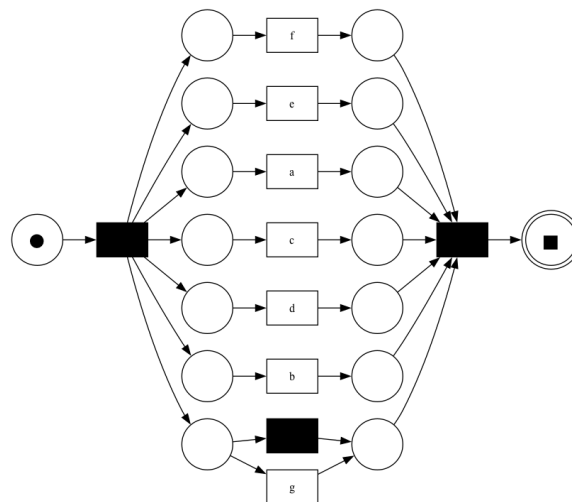
Item	Challenge	Log02	Log06	Log07	Mean	Question (short)
E-Sm-BQ-1	Behav. Quality	0	0	0	0.00	Does the model fix the internal ordering of the stable fragments?

Item	Challenge	Log02	Log06	Log07	Mean	Question (short)
E-Sm-BQ-2	Behav. Quality	1	1	n/a	1.00	Does the model allow fragment recombinations beyond the observed arrangement, with replay preserved?
E-Sm-IN-1	Interpret.	0	0	0	0.00	Are stable and flexible reference regions distinguishable?
E-Sm-IN-2	Interpret.	0	0	0	0.00	Is the output readable as one model spanning both regions?
Efit sum					1.00/8	= 12.5%

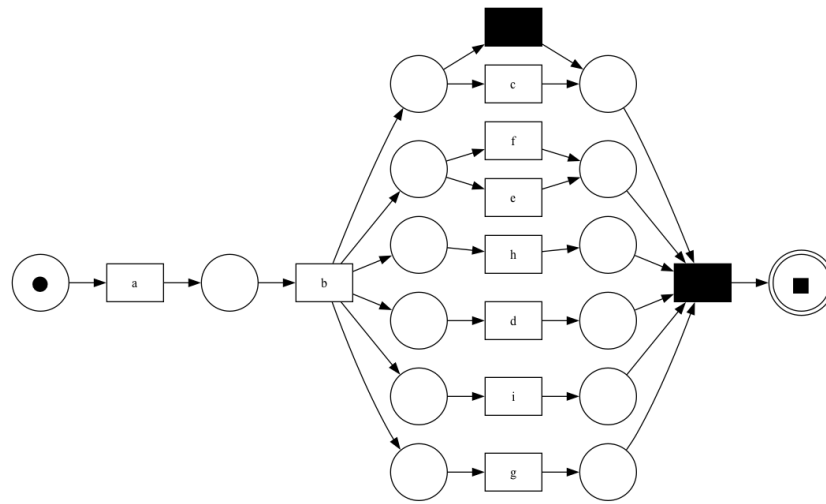
Rationale per item. On all three semi logs the IM places a flat AND block over nearly all activities (e.g. Log02: $+(a, c, X(\tau, f), b, X(d, e))$; Log06: $+(a, c, d, f, b, X(\tau, g), e)$). BQ-1: the flat AND block does not fix any internal fragment ordering — it allows arbitrary interleavings instead of fixed local ordering, precision 0.34–0.49 $\rightarrow$ 0. BQ-2: the observed variants are indeed replayable (fitness 1.0 is not misleading here), and the broad AND admits recombinations beyond the observed arrangement $\rightarrow$ 1 each (Log07 n/a). IN-1: no boundary, one broad block $\rightarrow$ 0. IN-2: no legible semantic zones, stable and flexible parts not separable as one model $\rightarrow$ 0.



Log02 - Semi-Structured - Petri net



Log06 - Semi-Structured - Petri net



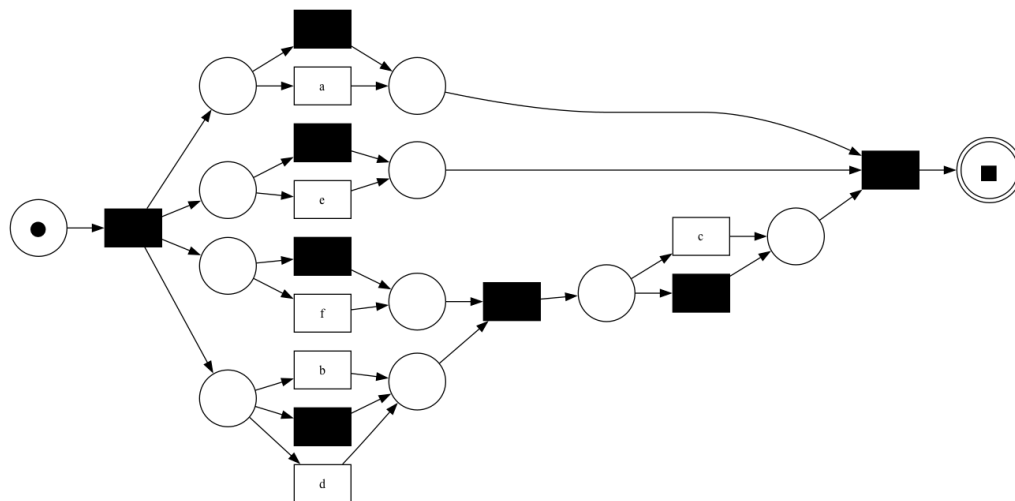
Log07 · Semi-Structured · Petri net

5 Class Loosely Structured — Efit = 29.2%

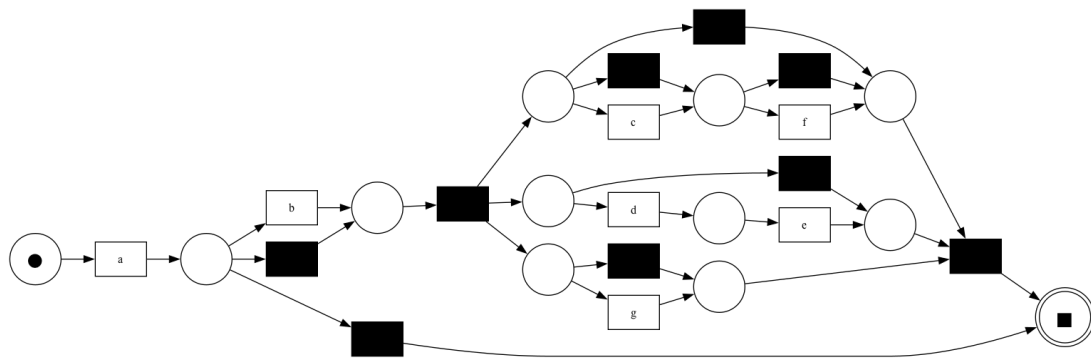
Logs: Log03, Log08, Log11. Behavioral Quality 41.7% · Interpretability 16.7%.

Item	Challenge	Log03	Log08	Log11	Mean	Question (short)
E-L-BQ-1	Behav. Quality	0	1	2	1.00	Does the model impose local restrictions against unsupported behavior?
E-L-BQ-2	Behav. Quality	0	1	1	0.67	Does the model allow behavior beyond the exactly observed variants, with replay preserved?
E-L-IN-1	Interpret.	0	0	0	0.00	Does the output condense to a few regularities?
E-L-IN-2	Interpret.	0	1	1	0.67	Local dependencies readable, without a global ordering?
Efit sum					2.33/8	= 29.2%

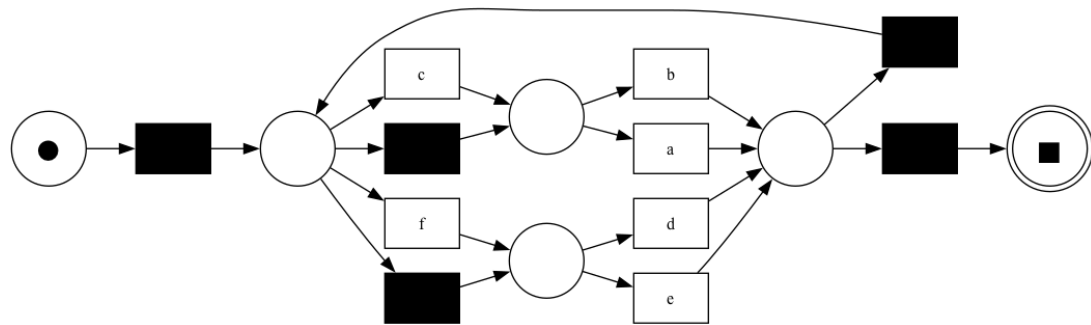
Rationale per item. BQ-1: inconsistent. Log11 realizes all three expected dependencies in the model $\rightarrow$ 2; Log08 partially (some bindings, but almost everything skippable via τ) $\rightarrow$ 1; Log03 barely restricts, every activity bypassable via $\tau \rightarrow$ 0. BQ-2: Log08/Log11 theoretically allow variants beyond the observed $\rightarrow$ 1; Log03 forces many orderings via the global ordering and allows runs with 0–1 activities $\rightarrow$ 0. IN-1: uniformly 0 — every model takes in every activity and inflates via τ (silent 5–9, CFC 10–13). IN-2: Log08/Log11 still partially show local dependencies or a freer loop ordering $\rightarrow$ 1; Log03 not readable $\rightarrow$ 0.



Log03 · Loosely Structured · Petri net



Log08 · Loosely Structured · Petri net



Log11 · Loosely Structured · Petri net

6 Comparison of the classes

Class	BQ	IN	Efit
Structured	91.7%	100.0%	95.8%
Semi-Structured	25.0%	0.0%	12.5%
Loosely Structured	41.7%	16.7%	29.2%

7 Classification

The empirical partial score of the Inductive Miner is not monotonic in structuredness: 95.8% → 12.5% → 29.2%. The collapse is in the middle, not at the loose pole.

Structured (95.8%): IM serves structured logs almost perfectly. Deterministic behavior is rendered fully replayable and tightly bound (precision 0.95–1.00), the models stay compact and left-to-right readable. Such logs are easy for any usable miner; the value is not very discriminating.

Semi-Structured (12.5%): the actual weakness. The class demands differential restriction — stable fragments tight, arrangement between them open. IM instead places a flat AND block over almost all activities, recognizes no fragments, and drives precision down to 0.34–0.49. The observed variants do remain replayable (BQ-2 = 1), but the internal fragment ordering, the region separation, and the readability are entirely missing — the other three items drop to 0.

Loosely Structured (29.2%): partial recovery on the behavioral side (BQ 41.7%), but weak interpretability (16.7%). Log11/Log08 partially realize local dependencies, Log03 barely. Throughout, the numerous τ -transitions inflate the models (IN-1 = 0), and the imperative Petri-net output forces a global ordering where local dependencies were called for.

Conclusion for the thesis. Unlike the Heuristics Miner (monotonic Efit gradient 91.7% → 70.8% → 25.0%), the Inductive Miner collapses hardest in the semi middle and places loosely (29.2%) above semi (12.5%). The class assignment does not follow from the absolute Efit alone, but from the full fit (Phase T + E) and the comparison with other miners. This assessment belongs, as an example, in Chapter 7.2 (Generalizability).